UTM
UNIVERSITI TEKNOLOGI MALAYSIA

02 (a – ii) TECHNIQUES IN AVOIDING AMBIGUITY: TEMPLATE-BASED

- *FORM TEMPLATES FOR QUALITY REQUIREMENTS*

Innovating Solutions

35

Form Template for Specifying Measurable Quality Requirements

FORM TEMPLATE: QUALITY REQUIREMENT

Item	Template	Example
ID	<Number of requirement>	R137.2
Goal	<Qualitatively stated goal>	Confirm room reservation immediately
Scale	<Scale for measuring the requirement>	1.Perfomance - Elapsed times in second (ratio scale) 2.Capacity - Number of concurrent users supported
Meter	<Procedure for measuring the requirement>	Timestamp start and end events; calculate time difference
Minimum	<Minimum acceptable quality to be achieved>	Less than 5s in at least 95% of all cases
OK range	<Value range that is OK and is aimed at>	Between 0.5 and 3s in more than 98% of all cases
Desired	<Quality achieved in the best possible case>	Less than 0.5 s in 100% of all cases

36

Form Template for Specifying Measurable Quality Requirements

FORM TEMPLATE: QUALITY REQUIREMENT

Item	Template	Requirements Example
ID	<Number of requirement>	**ID** / **Requirement Statement** R101.1 / The system **shall** display the dashboard within 2 seconds after successful user login
Goal	<Qualitatively stated goal>	**Goal** / **Requirement Statement** Display search results quickly / The system **shall** display the first batch of search results within 2 seconds after the user submits a query.
Scale	<Scale for measuring the requirement>	The system shall have an error rate of less than 0.1% during normal operations
Meter	<Procedure for measuring the requirement>	The system shall display the dashboard within 2 seconds after login
Minimum	<Minimum acceptable quality to be achieved>	The system **shall** have an operational error rate of **no more than 0.1%** during normal conditions
OK range	<Value range that is OK and is aimed at>	The system shall have an operational error rate between 0.05% and 0.1% during normal conditions.
Desired	<Quality achieved in the best possible case>	The system shall have an operational error rate of 0.01% or lower during normal conditions

37

UTM UNIVERSITI TEKNOLOGI MALAYSIA

02 (a – iii) TECHNIQUES IN AVOIDING AMBIGUITY: TEMPLATE-BASED

- *DOCUMENT TEMPLATES*

Innovating Solutions

38

Document Templates

TEMPLATE–BASED: DOCUMENT TEMPLATE

- Document templates help to systematically structure requirements document, for example a System Requirements Specification
- Example document template:
  - ISO/IEC/IEEE 29148:2018 SRS Template
  - Agile User Story–Driven Specification
  - Volere Requirements Specification Template
  - MIL-STD-961 (US Military Standard for system specifications)
- Often customers prescribe the use of a document template.

ISO/IEC/IEEE 29148:2018: High Level Structure

TEMPLATE–BASED: DOCUMENT TEMPLATE

1. Introduction
2.System Overview
3.Specific Requirements
4.Appendices

ISO/IEC/IEEE 29148:2018: SRS Structure

TEMPLATE-BASED: DOCUMENT TEMPLATE

1.Introduction	
1.1 Purpose	Purpose of the system
1.2 Scope	System name, high-level funtionality, key benefits, objectives and goals
1.3 References	A lis of all documents referenced in the document
1.4 Definitions	All words or phrases that have a special meaning used in the document
1.5 Overview	Give an overview of the content of this SRS document

41

ISO/IEC/IEEE 29148:2018: SRS Structure

TEMPLATE-BASED: DOCUMENT TEMPLATE

2. System Overview	
2.1 Product Perspective	Describes how the system fits within its environment, including relationships and dependencies with other systems.
2.2 System vision and goals	Explain the perspective of the software product here by describing the goal diagram
2.3 System context and boundary	Defines the scope of the system, what's inside and outside it, and its interactions with external entities.
2.4 Overall system structure	Provides a high-level view of the system's architecture, major components, and their relationships. Provide context diagram.
2.3 User characteristics	User groups that influence the requirements
2.4 Constraints	Restrictions of the system for example Regulation policies.

42

ISO/IEC/IEEE 29148:2018: SRS Structure

TEMPLATE–BASED: DOCUMENT TEMPLATE

3. Specific Requirements	
3.1 External Interface Requirements	Specifies how the system will interact with external systems, devices, and users.
3.1.1 User interface	Defines the appearance, layout, and interaction methods available to end users.
3.1.2 Hardware Interfaces	Describes connections and data exchange between the system and physical hardware devices.
3.1.3 Software Interfaces	Details integration points and data exchange between the system and other software applications or APIs.
3.1.4 Communication Interfaces	Specifies protocols, data formats, and communication methods used for information transfer.
3.2 Functional Requirements	Lists the specific capabilities and behaviors the system must perform. Can list all use cases
3.3 Use Case Descriptions	Documents detailed scenarios of user-system interactions, including all use case specifications.
3.4 Non-Functional Requirements	Defines quality attributes like performance, security, reliability, and usability.
3.5 Design Constraints	Outlines limitations and restrictions on system design, technology choices, or implementation.
3.6 Verification Requirements	Method of verifying each requirement (test, inspection)

Agile User Story–Driven Specification Template

TEMPLATE–BASED: DOCUMENT TEMPLATE

1. Introduction
2. References
3. User Story List
4. Detailed User Story Specifications
5. Non-Functional Requirements (Linked to Story)
6. Mockup / Wireframes
7. Traceability
8. Revision History

Agile User Story–Driven Specification Structure

TEMPLATE–BASED: DOCUMENT TEMPLATE

1.Introduction
- Briefly describe the purpose of this document and how it will be used (e.g., to capture and manage user stories with sufficient detail for implementation, testing, and traceability).

2. User Story List
- List all user story in a table

ID	User Story Name	Priority (Low / Medium / High)	Status (In Progress / Backlog / Done)

Agile User Story–Driven Specification Structure

TEMPLATE–BASED: DOCUMENT TEMPLATE

3. Detailed User Story Specifications
- Expands each user story with acceptance criteria, scenarios, and supporting details.

User Story ID:	ID of the story
User Story Name:	[Short name for the story]
User Story Statement:	As a [type of user], I want [an action or feature] so that [a benefit or reason].
Description / Notes:	• Brief elaboration of what the feature does, constraints, or context.
Acceptance Criteria (Given / When / Then format):	1.Given [precondition], When [action], Then [expected outcome]. 2.…
Priority:	High / Medium / Low
Status	Backlog / In Progress / Done
Dependencies	[List any prerequisite user stories or features]
Related Epic / Features	[Epic ID or name]

Agile User Story–Driven Specification Structure

TEMPLATE–BASED: DOCUMENT TEMPLATE

4. Non-Functional Requirements (Linked to Story)
   - Describes performance, security, usability, and other quality requirements tied to specific user stories.

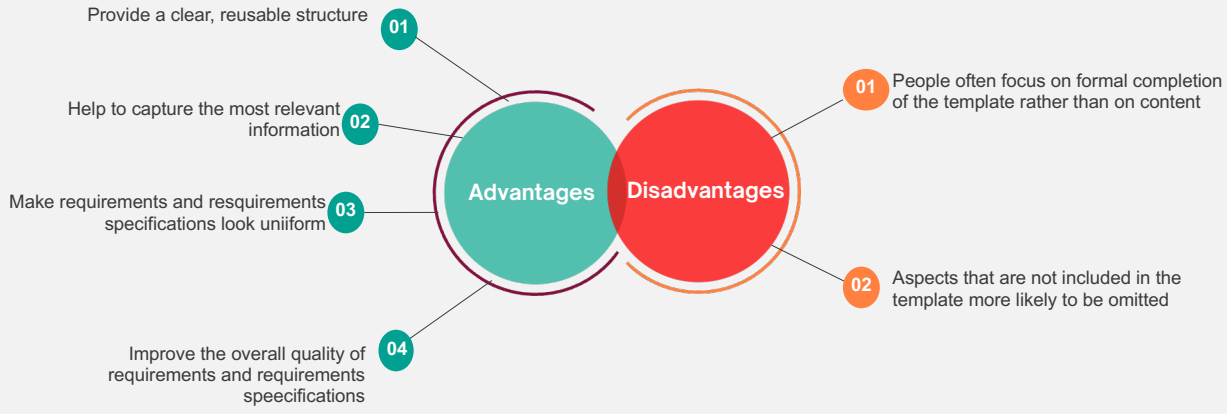
5. Mockup / Wireframes
   - Presents visual representations of screens, workflows, or layouts to support story understanding.
6. Traceability
   - Maps each user story to related requirements, design elements, and test cases to ensure coverage

7. Revision History
   - Tracks changes made to the document, including dates, descriptions, and authors.

47

Advantages & Disadvantages of Using Templates

TEMPLATE–BASED



48